Reconnaissance Activity Detection and Investigation using Sysmon and Wazuh

Objective

The objective of this simulation was to investigate how Sysmon and Wazuh detect Windows reconnaissance activity involving user enumeration, privilege discovery, group enumeration, network discovery, and process discovery techniques commonly associated with attacker situational awareness and pre-exploitation reconnaissance.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Discovery	System Owner/User Discovery	T1033
Discovery	Account Discovery	T1087
Discovery	Permission Groups Discovery	T1069
Discovery	System Network Configuration Discovery	T1016
Discovery	Process Discovery	T1057

Attack Simulation

The following reconnaissance commands were executed from an elevated PowerShell session:

```
whoami
```

```
whoami /priv
```

```
whoami /groups
```

```
net user
```

```
net localgroup administrators
```

```
ipconfig /all
```

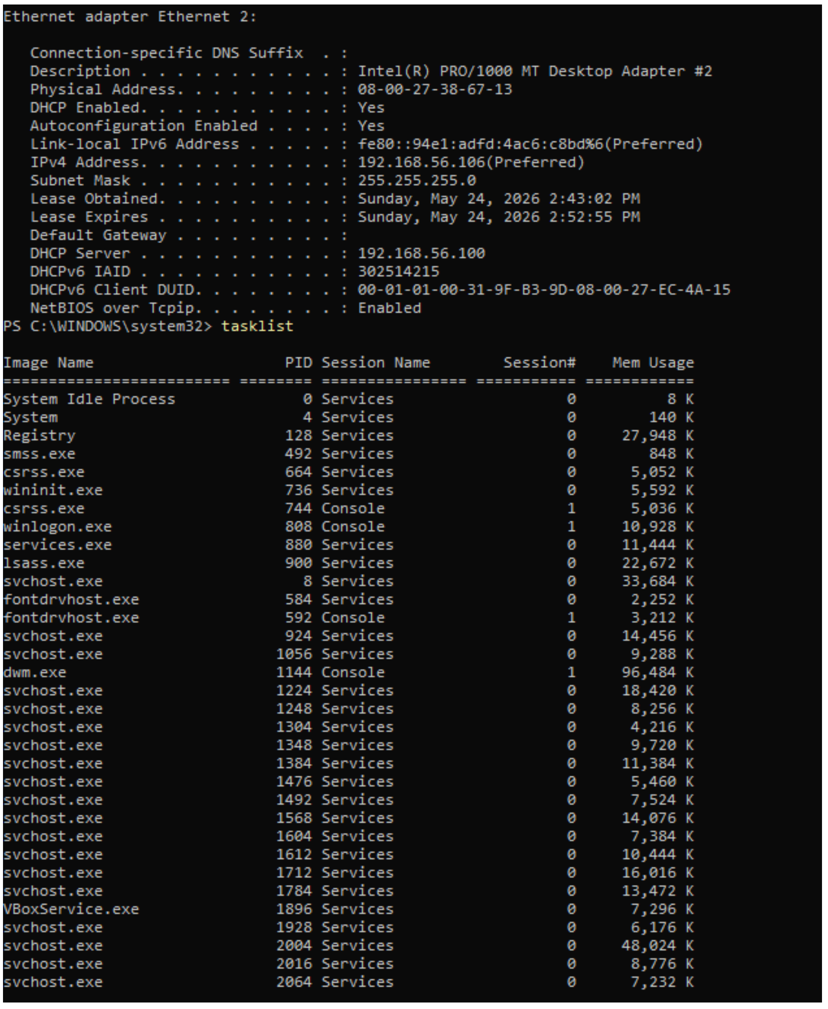
```
tasklist
```

Attack Chain Overview

Step	Activity
1	PowerShell launched reconnaissance commands
2	User and privilege enumeration performed
3	Local group membership queried
4	Network configuration discovered
5	Running process enumeration performed
6	Multiple child processes spawned from PowerShell

```
Select Administrator: Windows PowerShell
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> whoami
desktop-hdf09sq\socadmin
PS C:\WINDOWS\system32> whoami /priv

PRIVILEGES INFORMATION
----------------------

Privilege Name                            Description                                                        State
========================================= ================================================================== ========
SeIncreaseQuotaPrivilege                  Adjust memory quotas for a process                                 Disabled
SeSecurityPrivilege                       Manage auditing and security log                                   Disabled
SeTakeOwnershipPrivilege                  Take ownership of files or other objects                           Disabled
SeLoadDriverPrivilege                     Load and unload device drivers                                     Disabled
SeSystemProfilePrivilege                  Profile system performance                                         Disabled
SeSystemtimePrivilege                     Change the system time                                             Disabled
SeProfileSingleProcessPrivilege           Profile single process                                             Disabled
SeIncreaseBasePriorityPrivilege           Increase scheduling priority                                       Disabled
SeCreatePagefilePrivilege                 Create a pagefile                                                  Disabled
SeBackupPrivilege                         Back up files and directories                                      Disabled
SeRestorePrivilege                        Restore files and directories                                      Disabled
SeShutdownPrivilege                       Shut down the system                                               Disabled
SeDebugPrivilege                          Debug programs                                                     Enabled
SeSystemEnvironmentPrivilege              Modify firmware environment values                                 Disabled
SeChangeNotifyPrivilege                   Bypass traverse checking                                           Enabled
SeRemoteShutdownPrivilege                 Force shutdown from a remote system                                Disabled
SeUndockPrivilege                         Remove computer from docking station                               Disabled
SeManageVolumePrivilege                   Perform volume maintenance tasks                                   Disabled
SeImpersonatePrivilege                    Impersonate a client after authentication                          Enabled
SeCreateGlobalPrivilege                   Create global objects                                              Enabled
SeIncreaseWorkingSetPrivilege             Increase a process working set                                     Disabled
SeTimeZonePrivilege                       Change the time zone                                               Disabled
SeCreateSymbolicLinkPrivilege             Create symbolic links                                              Disabled
SeDelegateSessionUserImpersonatePrivilege Obtain an impersonation token for another user in the same session Disabled
PS C:\WINDOWS\system32> whoami /groups

GROUP INFORMATION
-----------------

Group Name                                                    Type             SID          Attributes
============================================================= ================ ============ ===============================================================
Everyone                                                      Well-known group S-1-1-0      Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account and member of Administrators group Well-known group S-1-5-114    Mandatory group, Enabled by default, Enabled group
BUILTIN\Administrators                                        Alias            S-1-5-32-544 Mandatory group, Enabled by default, Enabled group, Group owner
BUILTIN\Users                                                 Alias            S-1-5-32-545 Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                                      Well-known group S-1-5-4      Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON                                                 Well-known group S-1-2-1      Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users                              Well-known group S-1-5-11     Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization                                Well-known group S-1-5-15     Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account                                    Well-known group S-1-5-113    Mandatory group, Enabled by default, Enabled group
LOCAL                                                         Well-known group S-1-2-0      Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NTLM Authentication                              Well-known group S-1-5-64-10  Mandatory group, Enabled by default, Enabled group
Mandatory Label\High Mandatory Level                          Label            S-1-16-12288
```

```
PS C:\WINDOWS\system32> net user

User accounts for \\DESKTOP-HDF09SQ

-------------------------------------------------------------------------------
Administrator            DefaultAccount           Guest
socadmin                 WDAGUtilityAccount
The command completed successfully.

PS C:\WINDOWS\system32> net localgroup administrators
Alias name     administrators
Comment        Administrators have complete and unrestricted access to the computer/domain

Members

-------------------------------------------------------------------------------
Administrator
socadmin
The command completed successfully.

PS C:\WINDOWS\system32> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-HDF09SQ
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-EC-4A-15
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : fd17:625c:f037:2:fad8:9cb8:ccda:c957(Preferred)
   Temporary IPv6 Address. . . . . . : fd17:625c:f037:2:293c:6fa6:beb1:c8e0(Preferred)
   Link-local IPv6 Address . . . . . : fe80::950d:2704:ad74:aaf%3(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.0.2.15(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, May 24, 2026 10:40:14 AM
   Lease Expires . . . . . . . . . . : Monday, May 25, 2026 10:40:10 AM
   Default Gateway . . . . . . . . . : 10.0.2.2
   DHCP Server . . . . . . . . . . . : 10.0.2.2
   DHCPv6 IAID . . . . . . . . . . . : 235405351
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-9F-B3-9D-08-00-27-EC-4A-15
   DNS Servers . . . . . . . . . . . : 192.168.1.1
                                       114.114.114.114
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```



```
Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter #2
   Physical Address. . . . . . . . . : 08-00-27-38-67-13
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::94e1:adfd:4ac6:c8bd%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.106(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, May 24, 2026 2:43:02 PM
   Lease Expires . . . . . . . . . . : Sunday, May 24, 2026 2:52:55 PM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.56.100
   DHCPv6 IAID . . . . . . . . . . . : 302514215
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-9F-B3-9D-08-00-27-EC-4A-15
   NetBIOS over Tcpip. . . . . . . . : Enabled
PS C:\WINDOWS\system32> tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        140 K
Registry                       128 Services                   0     27,948 K
smss.exe                       492 Services                   0        848 K
csrss.exe                      664 Services                   0      5,052 K
wininit.exe                    736 Services                   0      5,592 K
csrss.exe                      744 Console                    1      5,036 K
winlogon.exe                   808 Console                    1     10,928 K
services.exe                   880 Services                   0     11,444 K
lsass.exe                      900 Services                   0     22,672 K
svchost.exe                      8 Services                   0     33,684 K
fontdrvhost.exe                584 Services                   0      2,252 K
fontdrvhost.exe                592 Console                    1      3,212 K
svchost.exe                    924 Services                   0     14,456 K
svchost.exe                   1056 Services                   0      9,288 K
dwm.exe                       1144 Console                    1     96,484 K
svchost.exe                   1224 Services                   0     18,420 K
svchost.exe                   1248 Services                   0      8,256 K
svchost.exe                   1304 Services                   0      4,216 K
svchost.exe                   1348 Services                   0      9,720 K
svchost.exe                   1384 Services                   0     11,384 K
svchost.exe                   1476 Services                   0      5,460 K
svchost.exe                   1492 Services                   0      7,524 K
svchost.exe                   1568 Services                   0     14,076 K
svchost.exe                   1604 Services                   0      7,384 K
svchost.exe                   1612 Services                   0     10,444 K
svchost.exe                   1712 Services                   0     16,016 K
svchost.exe                   1784 Services                   0     13,472 K
VBoxService.exe               1896 Services                   0      7,296 K
svchost.exe                   1928 Services                   0      6,176 K
svchost.exe                   2004 Services                   0     48,024 K
svchost.exe                   2016 Services                   0      8,776 K
svchost.exe                   2064 Services                   0      7,232 K
```

Figure 1 — Reconnaissance commands executed from PowerShell in the lab environment

Sysmon Investigation

Key Sysmon Events Identified

Sysmon Event ID	Detection Purpose
1	Process Creation

Primary Reconnaissance Utilities Observed

Process	Purpose
whoami.exe	User and privilege discovery
net.exe	Account and group discovery
net1.exe	Internal Windows networking utility
ipconfig.exe	Network configuration discovery
tasklist.exe	Process enumeration

Sysmon Event ID 1 — Reconnaissance Process Analysis

Key Findings

Field	Value
Event ID	1
Parent Process	powershell.exe
User	socadmin
Integrity Level	High
Activity Type	Discovery / Reconnaissance

Investigation Analysis

Sysmon Event ID 1 captured multiple reconnaissance utilities being launched from an elevated PowerShell session.

The observed process chain included:

```
powershell.exe
 ├── whoami.exe
 ├── net.exe
 │    └── net1.exe
 ├── ipconfig.exe
 └── tasklist.exe
```

This behavioral sequence strongly resembled attacker reconnaissance activity commonly observed during:

- initial access validation
- privilege assessment
- situational awareness
- lateral movement preparation

The parent-child relationship was particularly important because PowerShell spawning multiple reconnaissance utilities is significantly more suspicious than normal interactive command-line administration.

Important SOC Analyst Insight

Individually, these commands are:

- low severity
- frequently used by administrators

However, correlated execution of multiple reconnaissance utilities within a short timeframe significantly increases detection confidence.

Mature SOC investigations rely heavily on:

- behavioral correlation
- execution sequencing
- process ancestry
- user context

rather than isolated events.

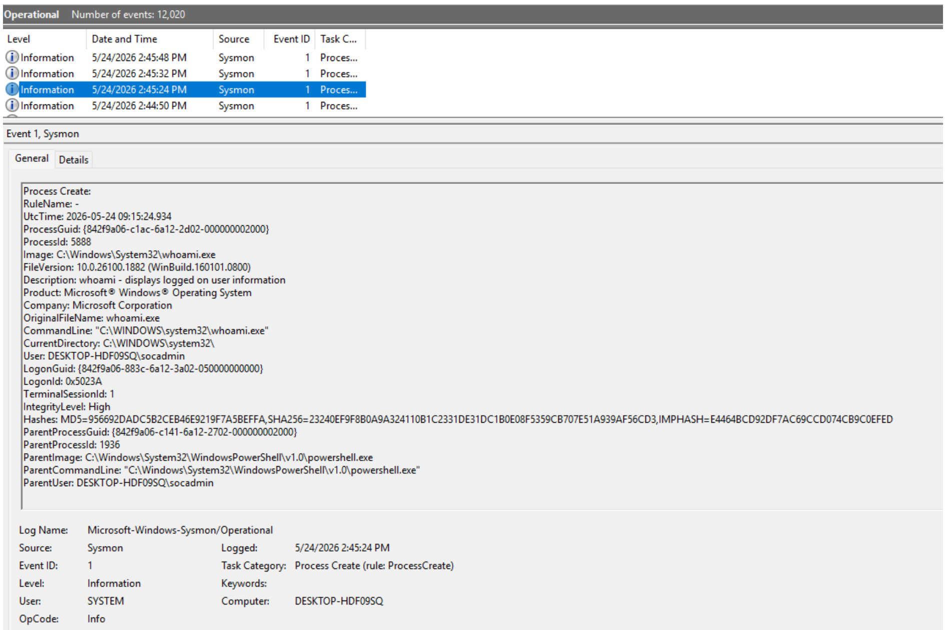


Figure 2 — Sysmon Event ID 1 showing PowerShell spawning whoami.exe reconnaissance activity

User and Privilege Discovery Analysis

whoami.exe Investigation

Commands Observed

```
whoami
whoami /priv
whoami /groups
```

Investigation Analysis

The `whoami.exe` utility was used to enumerate:

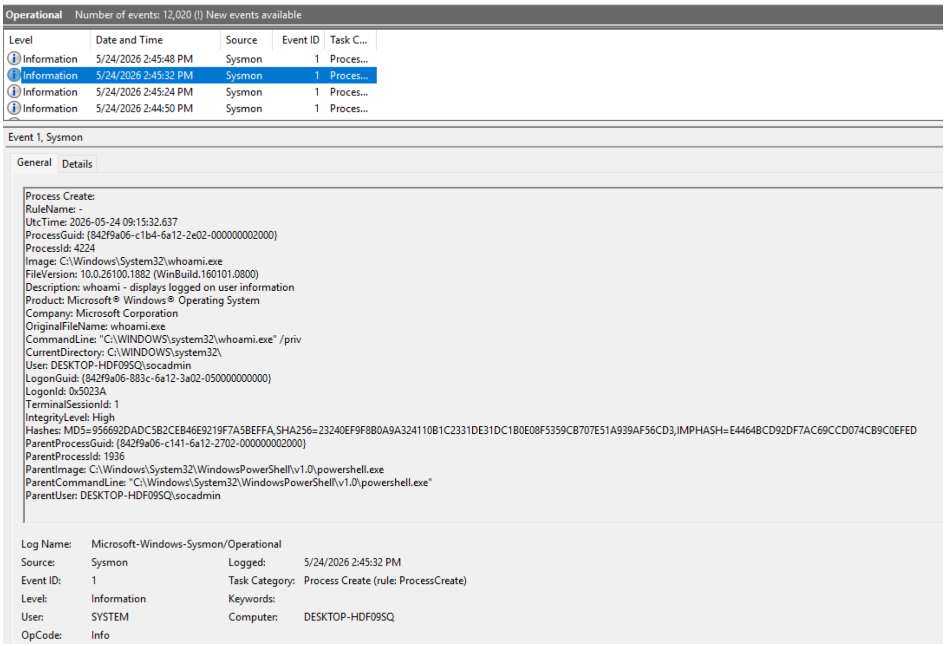
- current user context
- assigned privileges
- group memberships

This activity is commonly associated with:

- privilege discovery
- privilege escalation preparation
- environment awareness

The `/priv` and `/groups` arguments provided additional high-value telemetry because attackers frequently use these commands to identify:

- administrative privileges
- token privileges
- privileged group memberships
- escalation opportunities



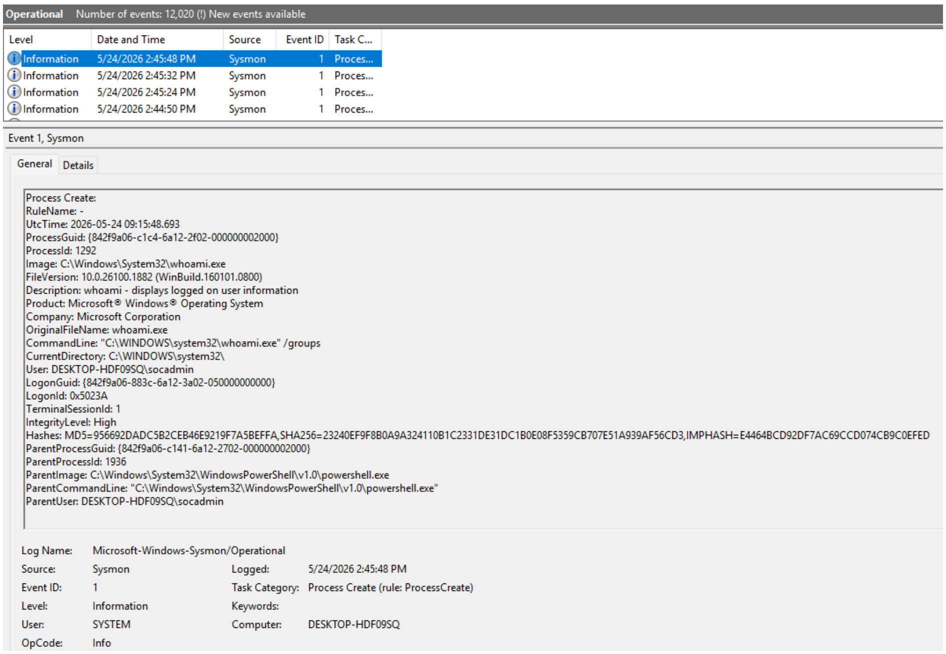


Figure 3 — Sysmon Event ID 1 showing privilege and group enumeration activity

Account and Group Enumeration Analysis

net.exe and net1.exe Investigation

Commands Observed

```
net user
net localgroup administrators
```

Investigation Analysis

The `net.exe` utility was used to enumerate:

- local user accounts
- administrative group membership

This activity commonly occurs during:

- account discovery
- privilege assessment
- lateral movement preparation

During execution, Windows internally launched:

```
net1.exe
```

This is legitimate Windows behavior and does not independently indicate malicious activity.

Understanding this relationship is important because junior analysts frequently misinterpret:

- net1.exe
- conhost.exe
- rundll32.exe

as malicious processes without proper context validation.

Important SOC Lesson

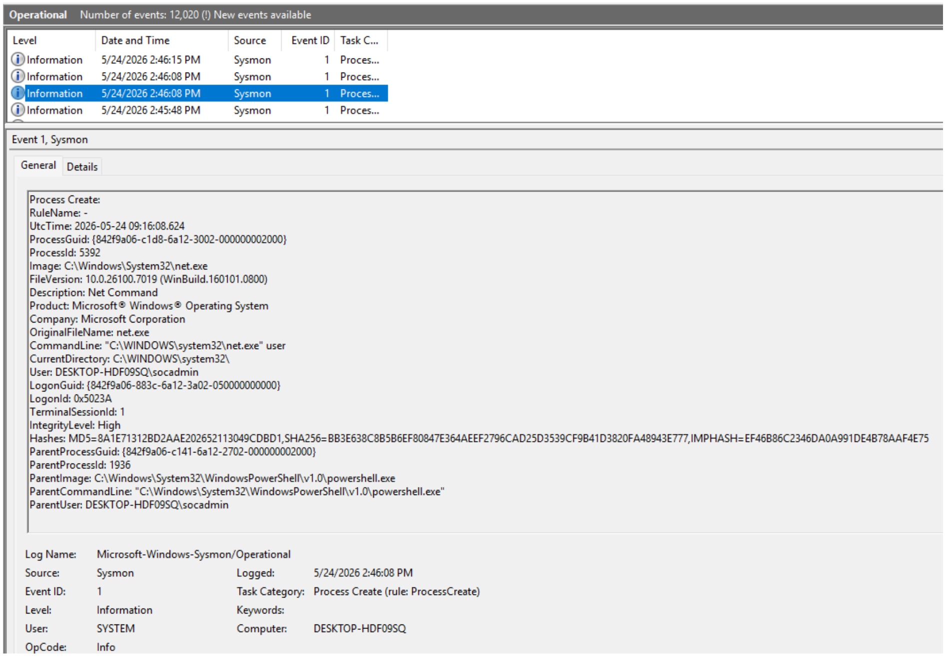
Context is more important than isolated events.

Detection engineering must distinguish between:

- legitimate administrative activity
- attacker reconnaissance behavior

The strongest indicator during this investigation was:

- multiple discovery commands
- executed sequentially
- launched from PowerShell
- under elevated context



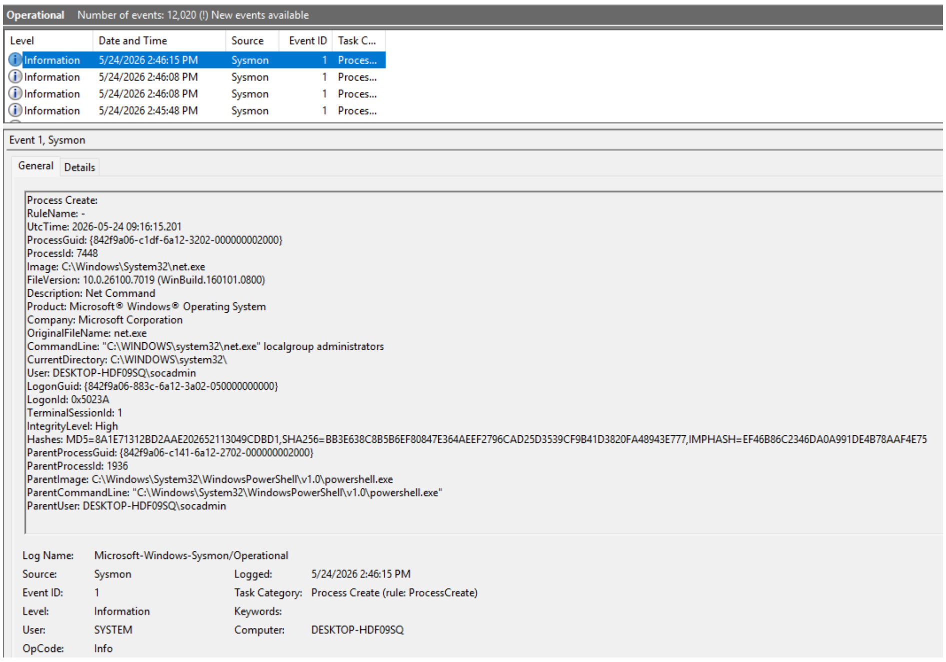


Figure 4 — Sysmon Event ID 1 showing account and administrator group enumeration

Network and Process Discovery Analysis

ipconfig.exe and tasklist.exe Investigation

Commands Observed

```
ipconfig /all
tasklist
```

Investigation Analysis

The `ipconfig.exe` utility was used to enumerate:

- IP addressing
- network adapters
- DNS configuration
- network connectivity information

This activity aligns with:

- System Network Configuration Discovery (T1016)

The `tasklist.exe` utility was used to enumerate:

- running processes
- active applications
- potential security tooling
- endpoint monitoring software

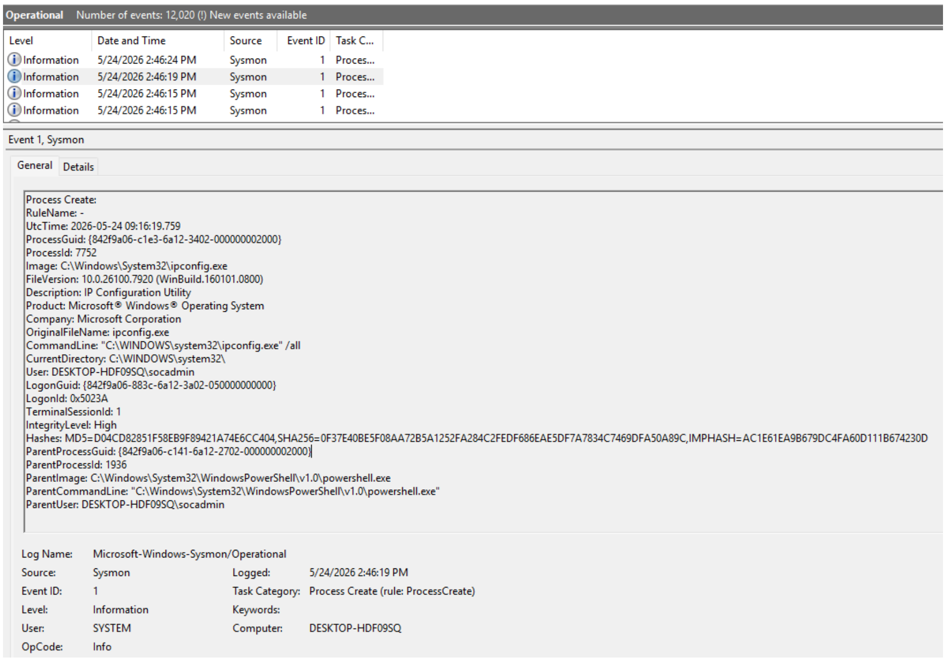
This activity aligns with:

- Process Discovery (T1057)

Attackers commonly use process discovery commands to identify:

- EDR software
- antivirus processes
- backup agents
- monitoring tools

prior to payload execution.



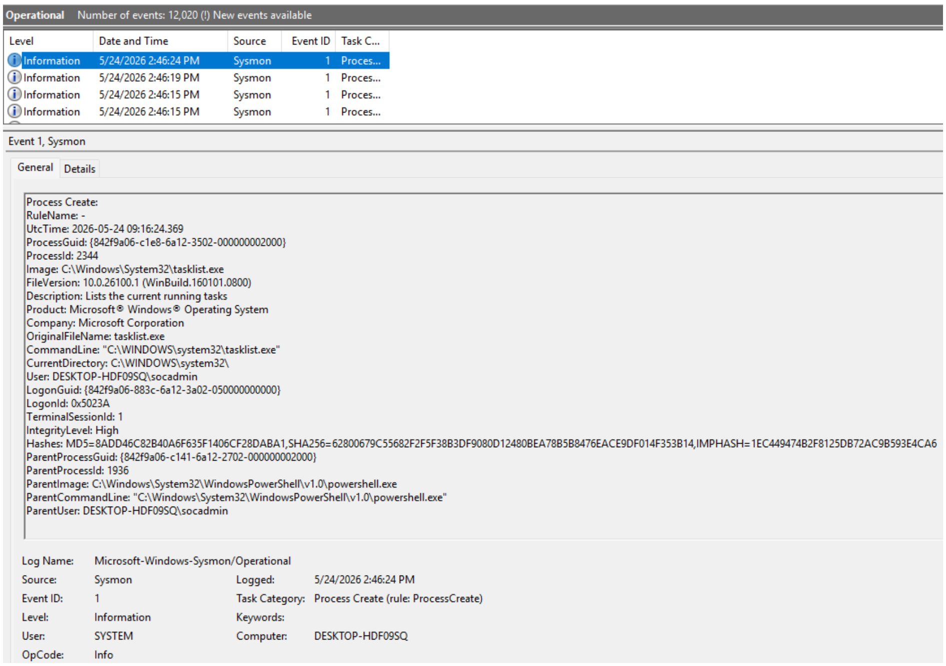


Figure 5 — Sysmon Event ID 1 showing network and process discovery activity

Wazuh Detection Analysis

Wazuh Detection Sources

Detection Source	Purpose
Sysmon Event Correlation	Reconnaissance process activity
Process Tree Analysis	PowerShell spawning discovery utilities
Behavioral Correlation	Sequential reconnaissance execution

Investigation Analysis

Wazuh successfully collected and correlated Sysmon telemetry associated with reconnaissance activity originating from PowerShell.

The strongest behavioral indicators included:

- PowerShell spawning multiple discovery utilities
- elevated execution context
- sequential reconnaissance execution
- process tree correlation
- account and privilege discovery behavior

The investigation demonstrated how:

- process ancestry
- command-line visibility
- behavioral sequencing

significantly improve detection fidelity during reconnaissance investigations.

Important False Positive Analysis

All commands used during this simulation:

- are legitimate Windows utilities
- are commonly used by administrators

However, the following factors significantly increased suspicion:

- multiple discovery commands executed together
- elevated PowerShell parent process
- rapid sequential execution
- broad environment enumeration behavior

This highlights the importance of behavioral analysis rather than relying solely on tool names.

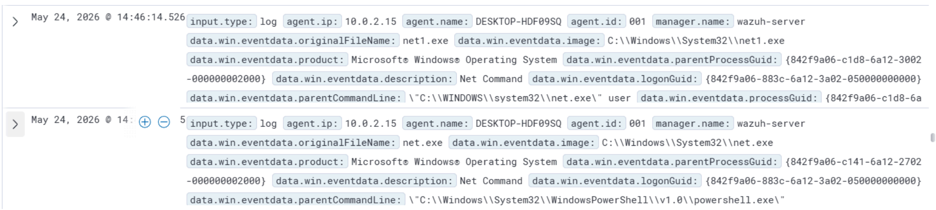


Figure 6 — Wazuh telemetry showing reconnaissance activity from PowerShell

{
"_index": "wazuh-alerts-4.x-2026.05.24",
"_id": "Ls5FWZ4Bsdig1CmudcSy",
"_version": 1,
"_score": null,
"_source": {
"input": {
"type": "log"
},
"agent": {
"ip": "10.0.2.15",
"name": "DESKTOP-HDF09SQ",
"id": "001"
},
"manager": {
"name": "wazuh-server"
},
"data": {
"win": {
"eventdata": {
"originalFileName": "net.exe",
"image": "C:\\\\Windows\\\\System32\\\\net.exe",
"product": "Microsoft® Windows® Operating System",
"parentProcessGuid": "{842f9a06-c141-6a12-2702-000000002000}",
"description": "Net Command",
"logonGuid": "{842f9a06-883c-6a12-3a02-050000000000}",
"parentCommandLine": "\\\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\"",
"processGuid": "{842f9a06-c1d8-6a12-3002-000000002000}",
"logonId": "0x5023a",
"parentProcessId": "1936",
"processId": "5392",
"currentDirectory": "C:\\\\WINDOWS\\\\system32\\\\",
"utcTime": "2026-05-24 09:16:08.624",
"hashes":
"MD5=8A1E71312BD2AAE202652113049CDBD1,SHA256=BB3E638C8B5B6EF80847E364AEEF2796CAD25D3539CF9B
"parentImage": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
"company": "Microsoft Corporation",

"commandLine": "\\\"C:\\\\WINDOWS\\\\system32\\\\net.exe\\\" user",
"integrityLevel": "High",
"fileVersion": "10.0.26100.7019 (WinBuild.160101.0800)",
"user": "DESKTOP-HDF09SQ\\\\socadmin",
"terminalSessionId": "1",
"parentUser": "DESKTOP-HDF09SQ\\\\socadmin"
},
"system": {
"eventID": "1",
"keywords": "0x8000000000000000",
"providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
"level": "4",
"channel": "Microsoft-Windows-Sysmon/Operational",
"opcode": "0",
"message": "\"Process Create:\r\nRuleName: -\r\nUtcTime: 2026-05-24 09:16:08.624\r\nProcessGuid: {842f9a06-c1d8-6a12-3002-000000002000}\r\nProcessId: 5392\r\nImage: C:\\Windows\\System32\\net.exe\r\nFileVersion: 10.0.26100.7019 (WinBuild.160101.0800)\r\nDescription: Net Command\r\nProduct: Microsoft® Windows® Operating System\r\nCompany: Microsoft Corporation\r\nOriginalFileName: net.exe\r\nCommandLine: \"C:\\WINDOWS\\system32\\net.exe\" user\r\nCurrentDirectory: C:\\WINDOWS\\system32\\\r\nUser: DESKTOP-HDF09SQ\\socadmin\r\nLogonGuid: {842f9a06-883c-6a12-3a02-050000000000}\r\nLogonId: 0x5023A\r\nTerminalSessionId: 1\r\nIntegrityLevel: High\r\nHashes:
MD5=8A1E71312BD2AAE202652113049CDBD1,SHA256=BB3E638C8B5B6EF80847E364AEEF2796CAD25D3539CF9B4
{842f9a06-c141-6a12-2702-000000002000}\r\nParentProcessId: 1936\r\nParentImage: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\r\nParentCommandLine: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" \r\nParentUser: DESKTOP-HDF09SQ\\socadmin\"",
"version": "5",
"systemTime": "2026-05-24T09:16:08.6319796Z",
"eventRecordID": "17273",
"threadID": "4088",
"computer": "DESKTOP-HDF09SQ",
"task": "1",
"processID": "3420",
"severityValue": "INFORMATION",
"providerName": "Microsoft-Windows-Sysmon"
}
}
},
"rule": {
"firedtimes": 1,
"mail": false,
"level": 3,
"description": "Discovery activity spawned via powershell execution",
"groups": [
"sysmon",
"sysmon_eid1_detections",
"windows"
],
"mitre": {
"technique": [
"Account Discovery",
"PowerShell"
],
"id": [
"T1087",
"T1059.001"
],
"tactic": [

"Discovery",
"Execution"
]
},
"id": "92033"
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779614174.2443559",
"timestamp": "2026-05-24T09:16:14.525+0000"
},
"fields": {
"timestamp": [
"2026-05-24T09:16:14.525Z"
]
},
"sort": [
1779614174525
]
}

{
"_index": "wazuh-alerts-4.x-2026.05.24",
"_id": "L85FWZ4Bsdig1CmudcSy",
"_version": 1,
"_score": null,
"_source": {
"input": {
"type": "log"
},
"agent": {
"ip": "10.0.2.15",
"name": "DESKTOP-HDF09SQ",
"id": "001"
},
"manager": {
"name": "wazuh-server"
},
"data": {
"win": {
"eventdata": {
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"image": "C:\\\\Windows\\\\System32\\\\net1.exe",
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"processId": "8500",
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"parentImage": "C:\\\\Windows\\\\System32\\\\net.exe",
"company": "Microsoft Corporation",
"commandLine": "C:\\\\WINDOWS\\\\system32\\\\net1 user",
"integrityLevel": "High",
"fileVersion": "10.0.26100.7019 (WinBuild.160101.0800)",
"user": "DESKTOP-HDF09SQ\\\\socadmin",
"terminalSessionId": "1",
"parentUser": "DESKTOP-HDF09SQ\\\\socadmin"
},
"system": {
"eventID": "1",
"keywords": "0x8000000000000000",
"providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
"level": "4",
"channel": "Microsoft-Windows-Sysmon/Operational",
"opcode": "0",
"message": "\"Process Create:\r\nRuleName: -\r\nUtcTime: 2026-05-24 09:16:08.663\r\nProcessGuid: {842f9a06-c1d8-6a12-3102-000000002000}\r\nProcessId: 8500\r\nImage: C:\\Windows\\System32\\net1.exe\r\nFileVersion: 10.0.26100.7019 (WinBuild.160101.0800)\r\nDescription: Net Command\r\nProduct: Microsoft® Windows® Operating System\r\nCompany: Microsoft Corporation\r\nOriginalFileName: net1.exe\r\nCommandLine: C:\\WINDOWS\\system32\\net1 user\r\nCurrentDirectory: C:\\WINDOWS\\system32\\\r\nUser: DESKTOP-HDF09SQ\\socadmin\r\nLogonGuid: {842f9a06-883c-6a12-3a02-050000000000}\r\nLogonId: 0x5023A\r\nTerminalSessionId: 1\r\nIntegrityLevel: High\r\nHashes: MD5=158DB954503C803C0840DD424FB608F8,SHA256=8489DB35375D890314EB8F7CD9E0CBD3866475F4E25286
{842f9a06-c1d8-6a12-3002-000000002000}\r\nParentProcessId: 5392\r\nParentImage: C:\\Windows\\System32\\net.exe\r\nParentCommandLine: \"C:\\WINDOWS\\system32\\net.exe\" user\r\nParentUser: DESKTOP-HDF09SQ\\socadmin\"",
"version": "5",
"systemTime": "2026-05-24T09:16:08.6707791Z",
"eventRecordID": "17274",
"threadID": "4088",
"computer": "DESKTOP-HDF09SQ",
"task": "1",
"processID": "3420",
"severityValue": "INFORMATION",
"providerName": "Microsoft-Windows-Sysmon"
}
}
},
"rule": {
"firedtimes": 1,
"mail": false,
"level": 3,
"description": "Discovery activity executed",
"groups": [
"sysmon",
"sysmon_eid1_detections",
"windows"
],
"mitre": {
"technique": [
"Account Discovery"
],
"id": [
"T1087"
],
"tactic": [
"Discovery"

]
},
"id": "92031"
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779614174.2449302",
"timestamp": "2026-05-24T09:16:14.526+0000"
},
"fields": {
"timestamp": [
"2026-05-24T09:16:14.526Z"
]
},
"sort": [
1779614174526
]
}

Detailed Wazuh telemetry for reconnaissance process execution

---

Important Telemetry Validation

Additional registry inventory-related events generated by:

- svchost.exe
- Windows inventory services
- telemetry bookkeeping

were observed during investigation.

Examples included:

- InvDB-Path
- InvDB-Pub
- InvDB-Ver

These events were determined to be:

- legitimate Windows telemetry activity
- non-malicious background operations

This reinforced an important SOC principle:

```
Context is more important than raw event volume.
```

Attack Timeline

Time	Event
2:45 PM	PowerShell launched reconnaissance commands
2:45:24 PM	whoami.exe executed
2:45:32 PM	whoami /priv executed
2:45:48 PM	whoami /groups executed
2:46:08 PM	net.exe executed
2:46:15 PM	net1.exe spawned internally

Time	Event
2:46:19 PM	ipconfig.exe executed
2:46:24 PM	tasklist.exe executed
2:46:14PM	Wazuh collected correlated reconnaissance telemetry

(Replace with your actual timestamps from Event Viewer)

Key Detection Findings

Finding	Impact
PowerShell spawning recon tools	Increased behavioral suspicion
User and privilege discovery	Privilege assessment behavior
Local group enumeration	Administrative discovery activity
Network configuration discovery	Environment awareness activity
Process enumeration	Security tooling discovery
Sequential command execution	Improved detection confidence
Parent-child process correlation	High-value behavioral telemetry

Primary Investigation Artifacts

Sysmon Event IDs

Event ID	Meaning
1	Process Creation

Reconnaissance Utilities Observed

Utility	ATT&CK Mapping
whoami.exe	T1033
whoami /priv	Privilege Discovery
whoami /groups	Group Discovery
net user	T1087
net localgroup administrators	T1069
ipconfig /all	T1016
tasklist	T1057

Conclusion

This simulation demonstrated how Sysmon and Wazuh provide strong visibility into Windows reconnaissance activity commonly associated with attacker situational awareness and pre-exploitation discovery behavior.

Behavioral correlation between:

- PowerShell execution
- privilege discovery
- account enumeration
- network discovery
- process enumeration
- parent-child process relationships

significantly improved detection confidence and investigation quality.

The investigation also highlighted the importance of contextual analysis because many reconnaissance utilities are legitimate administrative tools whose maliciousness depends heavily on:

- execution sequence
- process ancestry
- timing
- user context
- behavioral correlation

rather than the commands themselves alone.